MIL-F-42012A
9 January 1987
SUPERSEDING
MIL-F-42012(DM)
29 July 1986

MILITARY SPECIFICATION

FORCEPS, IRIS (Graefe)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one size of iris forceps of the Graefe design with one-by-two teeth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6515
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:10Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

GG-I-526 - Instruments, Dental and Surgical; General Specification.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-up.

MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARDS

FEDERAL

Federal Test Method Standard No. 151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of solicitation.

DPSC Drawing 23381 - FORCEPS, IRIS, GRAEFE

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Forceps shall be manufactured from Class 4 martensitic corrosion-resistant steel of composition in accordance with ASTM F899 for the class and type used. Composition shall be tested as specified in 4.5.1.

3.2 Style and design. Forceps shall be in accordance with DPSC drawing 23381. Teeth shall mesh uniformly when the forceps are closed with no lateral pressure.

3.3 Passivation. Forceps shall be properly passivated.

3.4 Hardness. Forceps shall have a Rockwell hardness of HRC 40-48. Hardness shall be tested as specified in 4.5.2.

3.5 Finish. The forceps shall have a mirror finish as defined in GG-I-526. The finish shall be uniform and free from sharp edges, grind or tool marks, burrs, chips, cracks, pitting, and crevices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

3.6 Resistance to Corrosion. Forceps shall show no signs of corrosion or plating of copper sulfate when tested as specified in 4.5.3.

3.7 Markings. Forceps shall be permanently marked with the manufacturer's name or registered trademark and the word "stainless." In addition, forceps of foreign manufacture shall bear the name of the country of manufacture.

3.8 Workmanship. The forceps shall be clean and free from any defect which affects their performance or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, or does it commit the Government to acceptance of defective material.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc. investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties), of the item.

4.2.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

Source: https://assist.dla.mil -- Downloaded: 202640-21T20:10Z
Check the source to verify that this is the current version before use.

4.2.2 Certificate of quality. Certificate of quality supplied by the manufacturer of the steel may be furnished in lieu of actual performance of composition testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number, the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.3 Sampling.

4.3.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purpose shall be one forceps.

TABLE I. Sampling criteria for examination

Type of defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.3.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purposes shall be one forceps. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling criteria for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Composition	3.1	4.5.1	1/	2/
Dimensions	3.2	---	S-2	2.5
Hardness	3.4	4.5.2	S-2	1.0
Corrosion	3.6	4.5.3	S-2	1.0

1/ A sufficient amount of raw material shall be tested.

2/ The lot shall be rejected if test requirements are not met.

4.4 Examination.

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:10Z
Check the source to verify that this is the current version before use.

4.4.1 Examination of forceps. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any forceps in the sample containing one or more defects shall be rejected and if the number of defective forceps in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	Teeth do not mesh completely or uniformly without lateral pressure.
102	Forceps not free from cracks, pits, burrs, sharp edges, scale, rough areas or corrosion.
103	Finish of forceps not uniform or not as specified.
104	Forceps has grind or tool parts.
105	Identification markings not present, not complete or not legible.
Minor	
201	Forceps not free from dirt, polishing material or foreign matter.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Composition. Raw material shall be tested for composition in accordance with Federal Test Method 111 or 112 of Federal Test Method Standard Number 151 (see 4.2.2).

4.5.2 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.5.3 Corrosion tests. Forceps shall be tested for resistance to corrosion as specified in 3.6 and 4.5.3.1 and 4.5.3.2.

4.5.3.1 Boil test. The forceps shall be tested in accordance with the resistance to boil test of federal specification GG-I-526.

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:10Z
Check the source to verify that this is the current version before use.

4.5.3.2 Copper sulfate test. The forceps shall be tested in accordance with the resistance to copper sulfate test of federal specification GG-I-526.

4.6 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Each forceps shall be cleaned and dried in accordance with MIL-P-116.

5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit Package. Each forceps shall be packaged in accordance with MIL-P-116, method IC-2. As an alternate, each forceps shall be packaged in a rigid plastic container having a friction or plug-type closure. The forceps shall be immobilized within the container and the operative end shall be protected with suitable cushioning material.

5.1.1.4 Intermediate package. Twelve unit packages shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for each forceps shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Twelve unit packges shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202640-21T20:10Z
Check the source to verify that this is the currentversion before use.

5.2.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure,strapping and waterproofing shall be as specified in the box specification.

5.2.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. Twelve intermediate packages (144 units) shall be packed in accordance with the requirements of MIL-STD-794 for level C.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A, B. Forceps packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Forceps packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3. Marking.

5.3.1 Forceps. Each forceps shall be permanently marked as specified in 3.7.

5.3.2 Levels A, B, and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.

6. NOTES

6.1 Intended use. The forceps are intended for gripping the iris during surgery of the eye.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.3 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-01-139-8280	FORCEPS, IRIS, Graefe, Straight, 2-1/2 Inch

6.4 Subject term (key word) listing.

Forceps, Eye
Graefe Forceps

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3917

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide *suggestions.* This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Personnel Support Center
ATTN: DPSC-ATTS
Philadelphia, PA 19101

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:10Z
Check the source to verify that this is the current version before use.